



**AI-DRIVEN PREDICTIVE MAINTENANCE SYSTEM FOR
ROTATING MACHINERY USING MULTISENSORY DATA**

A PROJECT REPORT

Submitted by

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in the partial fulfillment of the award for the degree

of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

P.S.R.R. COLLEGE OF ENGINEERING

SIVAKASI 626-140

ANNA UNIVERSITY, CHENNAI-600 025

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BONAFIDE CERTIFICATE

Certified that this project report “**AI-DRIVEN PREDICTIVE MAINTENANCE SYSTEM FOR ROTATING MACHINERY USING MULTISENSORY DATA**”, is the Bonafide work of **BAL ABISHEAK C (952022243010)**, **FRANCIES T (952022243016)**, **VINAY PRASATH R (952022243049)** who carried out project work under my supervision.

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ABSTRACT

Rotating machinery such as motors, pumps, compressors, and turbines plays a vital role in modern industrial operations. Unexpected failures in these systems can lead to significant downtime, financial losses, and safety risks. Traditional maintenance strategies, including reactive and time-based preventive methods, are often inefficient in handling real-time machine health conditions. This project presents the design and implementation of an AI-driven predictive maintenance system for rotating machinery using multisensory data. The system utilizes an Arduino UNO microcontroller integrated with multiple sensors to monitor key operational parameters. These include vibration and tilt detection using the MPU6050 sensor, temperature monitoring using the LM35 sensor, current measurement using the ACS712 sensor, voltage monitoring through a voltage sensor module, and acoustic anomaly detection using a sound sensor. The acquired sensor data is processed using a rule-based fault detection approach, where each parameter is evaluated against predefined threshold values to identify abnormal conditions. The system classifies machine states into normal operation or specific fault categories such as tilt, overheating, noise anomaly, electrical overload, and voltage fluctuation. The results are displayed in real time on a 16x2 LCD and transmitted to a cloud platform via the ESP8266 WiFi module for remote monitoring and data logging. The system demonstrates fast response time and reliable fault detection under different operating conditions. The multisensory approach improves accuracy compared to single-sensor systems, while IoT integration enables continuous monitoring and analysis. This work provides a cost-effective and scalable solution for predictive maintenance in small and medium-scale industrial applications and establishes a foundation for future integration with machine learning based analytics.

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LIST OF ABBREVIATIONS

ABBREVIATION		DESCRIPTION
AI	-	Artificial Intelligence
IoT	-	Internet of Things
ML	-	Machine Learning
MCU	-	Microcontroller Unit
ADC	-	Analog to Digital Converter
GPIO	-	General Purpose Input/Output
I2C	-	Inter Integrated Circuit
UART	-	Universal Asynchronous Receiver Transmitter
LCD	-	Liquid Crystal Display
MPU	-	Motion Processing Unit
PWM	-	Pulse Width Modulation
RF	-	Random Forest
SVM	-	Support Vector Machine
RMS	-	Root Mean Square
HTTP	-	Hypertext Transfer Protocol
FFT	-	Fast Fourier Transform
PCB	-	Printed Circuit Board
IDE	-	Integrated Development Environment
SDK	-	Software Development Kit